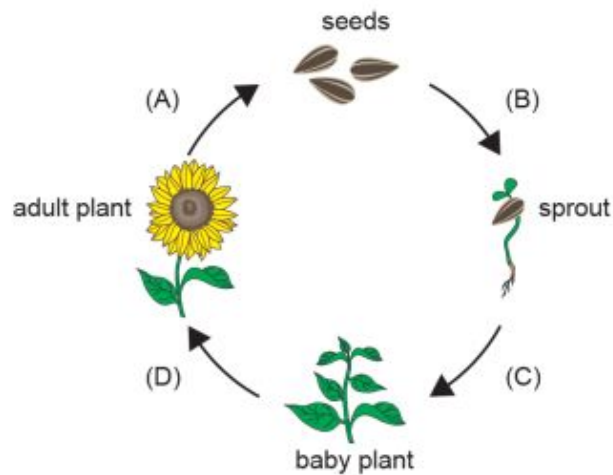


ICAS Paper C Science

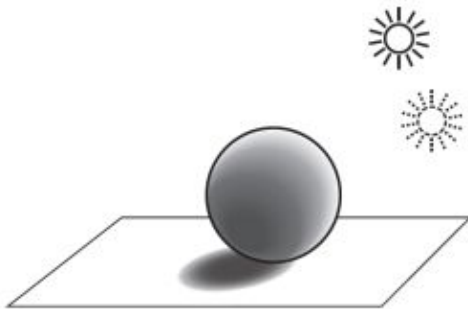
Year 2018

Q 1. Pollination is part of the process in which a flower produces seeds. The diagram shows the life cycle of a plant.

At which stage does pollination take place according to the diagram?



Q 2. The drawing shows the Sun shining on a ball on a sheet of paper.



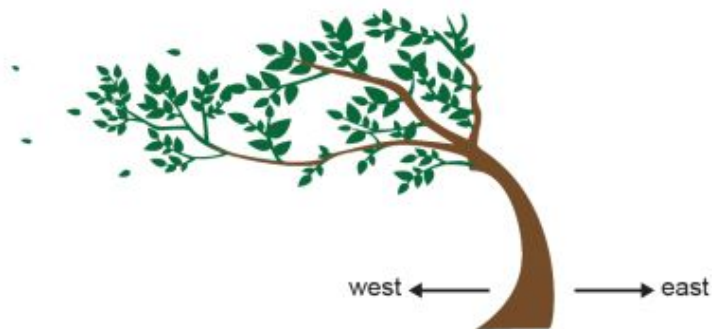
What will happen to the shadow of the ball as the Sun moves to the dotted position?

The shadow of the ball will

- | | |
|---------------------|---------------------|
| (A) become longer. | (B) become shorter. |
| (C) become rounder. | (D) disappear. |

Q 3. An observation is anything you can see, hear, feel, smell or taste.

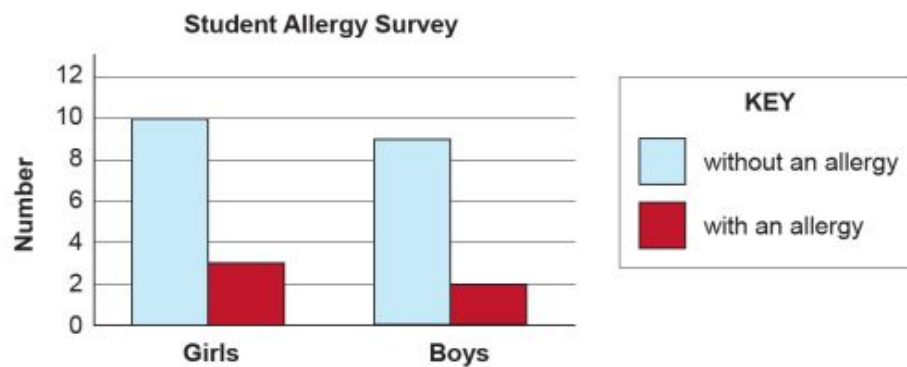
On one windy day, Bob notices this tree in front of his house.



Which of these is an observation Bob can make?

- (A) It is going to get cold.
- (B) The tree branches may break.
- (C) The wind is blowing to the east.
- (D) The tree branches are bending to the west.

Q 4. The graph shows the results of a student allergy survey in one class.



Which of the following groups has the greatest number of students in it?

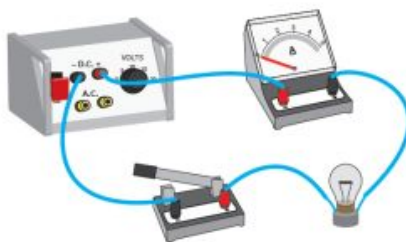
- (A) students with an allergy
- (B) girls with an allergy
- (C) boys without an allergy
- (D) girls without an allergy

Q 5. We use circuit symbols as an easy way to draw electrical circuits.

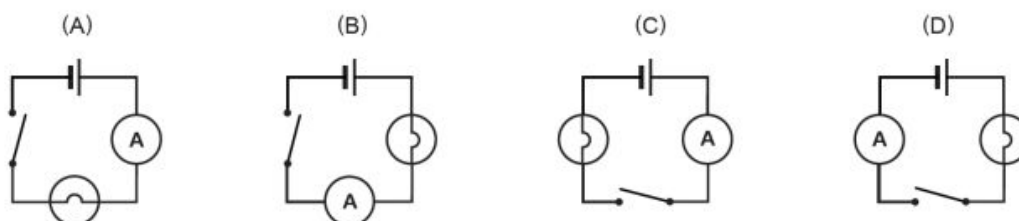
The diagram shows the symbols for some common electrical parts.

Part				
Symbol				

Amy made this circuit by joining some electrical parts together.

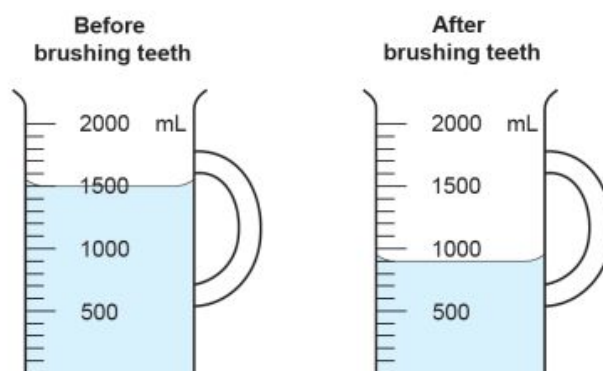


Which diagram represents Amy's circuit?



Q 6. Daniel often leaves the tap running while he brushes his teeth. His parents asked him to find out how much water he actually needs to brush his teeth.

Daniel brushed his teeth by using sips of water from a jug.



How many millilitres (mL) of water did Daniel use to brush his teeth?

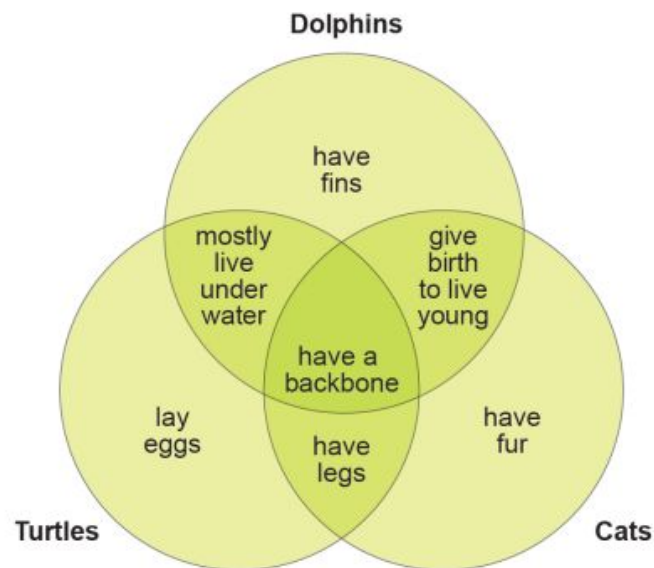
- (A) 1500

(C) 600

(B) 900

(D) 400

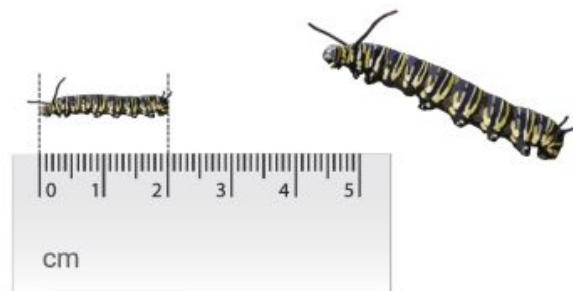
Q 7. The diagram shows some features of three animals.



Which feature is common to dolphins and cats but not to turtles?

- (A) They lay eggs. (B) They have a backbone.
(C) They give birth to live young. (D) They mostly live under water.

Q 8. Wendy found two caterpillars crawling in her yard. She put a ruler next to one caterpillar, as shown.



How many centimetres long is the other caterpillar?

- (A) 5 (B) 4 (C) 3 (D) 2

Q 9. Ned read that when water freezes it expands and takes up more space. He poured a glass of water before freezing it.



What should he do to investigate whether the statement is correct?

- (A) mark the water level before and after freezing
(B) weigh the glass of water before and after freezing
(C) compare the volume of water with the mass of the ice
(D) measure the water temperature before and after freezing

- Q 10. Objects around us are made of different materials. Each material has properties that make it suitable for a particular use.

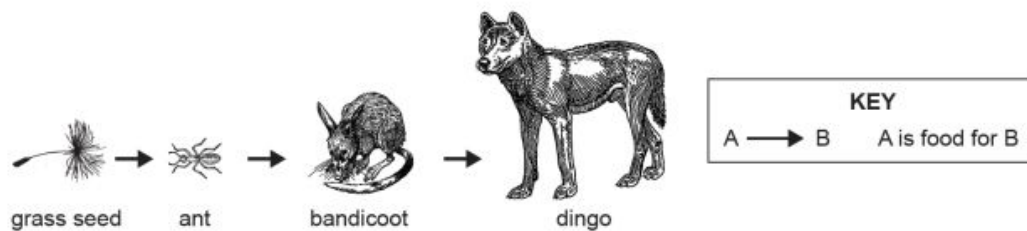
The picture shows a bucket made of metal.



What property of this material makes it suitable for a bucket?

- (A) how big it is
(B) whether it is waterproof
(C) whether it is shiny
(D) how round it is
- Q 11. A food chain shows how nutrients and energy in food are passed on from one living thing to the next.

The diagram shows part of a food chain.



Due to the spread of a disease, the number of bandicoots decreased rapidly.

What is likely to happen as a result?

- (A) The number of dingoes and the number of ants will both increase.
(B) The number of dingoes will decrease and the number of ants will increase.
(C) The number of dingoes will increase and the number of ants will decrease.
(D) The number of dingoes and the number of ants will both decrease.
- Q 12. Sally uses this padlock to chain her bicycle to a bike rack.



To unlock the padlock she has to carry out the following steps:

1. Turn the dial to zero using the shortest distance.
2. Turn the dial anticlockwise and stop at 30.
3. Turn the dial clockwise and stop at 25.
4. Turn the dial back to 0 using the shortest distance.

How many times will the number 20 go past the red arrow from its current position?

- (A) 1
(B) 2
(C) 3
(D) 4

Q 13. The table lists some components of 100 grams (g) of the milk that some mammals produce.

Mammal	Protein (g)	Fat (g)	Lactose (g)	Energy (kcal)
cow	3.3	3.9	4.6	66
human	1.3	4.1	7.2	69
goat	3.1	3.7	4.4	62
sheep	5.4	5.8	5.1	93

Which of these mammals produces the milk with the highest energy content?

- (A) cow (B) human (C) goat (D) sheep

Q 14. Which statement is supported by the data in the table?

- (A) The greater the protein content of the milk, the higher the energy content of the milk.
 (B) The greater the fat content of the milk, the higher the energy content of the milk.
 (C) In each type of milk, the lactose content is higher than the protein or fat content.
 (D) Human milk has the lowest fat content of the four types of milk listed.

Q 15. Animals that have a backbone are called vertebrates.

The table shows five groups of vertebrates and their characteristics.

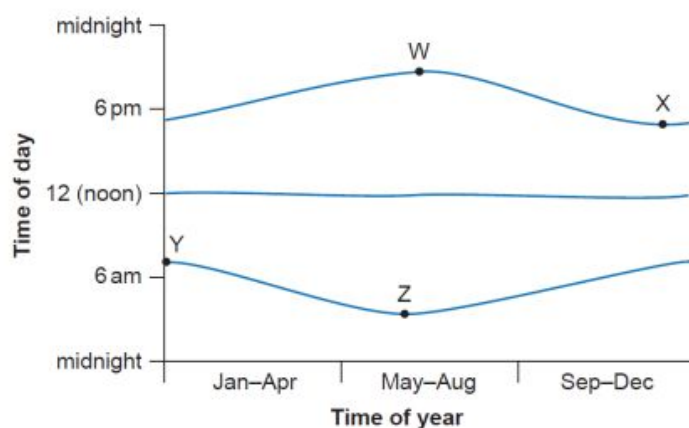
	Fish	Amphibians	Reptiles	Birds	Mammals
Form of reproduction	lay eggs with no outer shell	lay eggs with no outer shell	lay eggs with outer shell	lay eggs with outer shell	give birth to live young
Body covering	interlocking scales	moist skin	interlocking scales	feathers	dry skin with hair
Method of breathing	through gills	through skin or gills; lungs as adult	through lungs	through lungs	through lungs
Cold- or warm-blooded	cold-blooded	cold-blooded	cold-blooded	warm-blooded	warm-blooded

Tuatara are animals that have interlocking scales and breathe through lungs.

Which statement about tuatara is true?

- (A) They lay eggs with no outer shell.
 (B) They are fish.
 (C) They are reptiles.
 (D) They are warm-blooded.

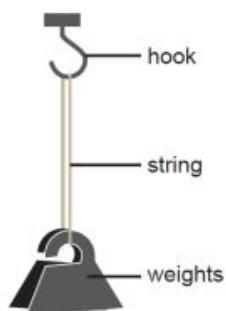
Q 16. The graph shows the times of sunrise and sunset at one location on Earth over a year.



Which point represents the earliest sunset?

- (A) W (B) X (C) Y (D) Z

Q 17. A group of students wanted to compare the strength of two strings made of different materials, X and Y. The diagram shows how they set up their experiment.



The students kept adding weights until the string made of material X broke.

They then repeated the experiment with the string made of material Y.

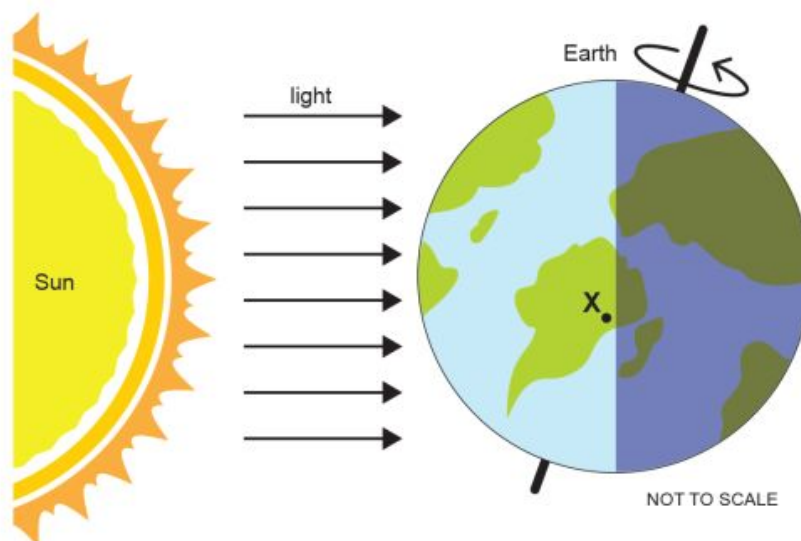
Which of these do they have to measure to answer their investigation question?

- (A) the length of each string (B) the time taken for each string to break
(C) the number of weights each string could hold (D) the amount of stretch in each string

Q 18. What do the students need to keep the same to make this a fair test?

- (A) the number of tests repeated for each string
(B) the material of each string
(C) the number of weights hanging from each string
(D) the length of each string

- Q 19. As Earth orbits the Sun, it also spins on its axis. It takes about 24 hours for one complete spin. It is daytime on the side of Earth facing the Sun and it is night-time on the side facing away from the Sun.

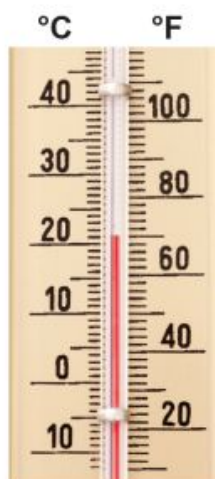


What is the time of day at the location marked X on the diagram?

- (A) midday (B) midnight (C) sunrise (D) sunset

- Q 20. Some thermometers show temperature in two units: degrees Celsius ($^{\circ}\text{C}$) and degrees Fahrenheit ($^{\circ}\text{F}$).

The photograph shows part of a thermometer with these two units.



What is the temperature in degrees Fahrenheit ($^{\circ}\text{F}$)?

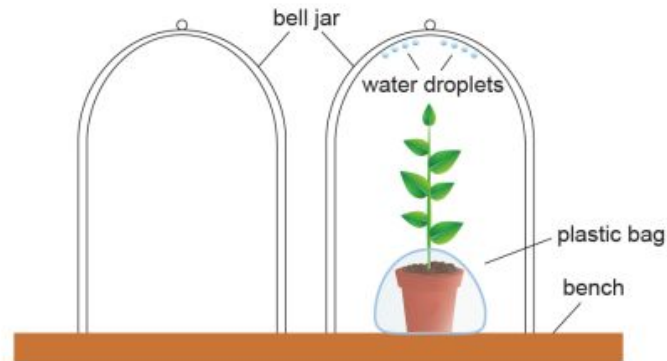
- (A) 21
(B) 65
(C) 70
(D) 85

Q 21. Liam's teacher told him that plants give off water vapour.

He set up two sealed bell jars. One had a plant in a pot with damp soil in it, the soil covered with a plastic bag.

After 24 hours, he found that the bell jar containing the plant had water droplets on the inside.

Liam compared the bell jar with the plant in it, to an empty bell jar.



Why did Liam have to cover the soil in the pot?

- (A) to stop the plant from drying out overnight
- (B) so that the bell jar would not smell like soil
- (C) to stop water from the soil interfering with the results
- (D) so that the droplets on the bell jar could not fall onto the soil

Q 22. The photo shows a small collection of shells. The key helps determine the type of a shell.



Key to Shells		
i)	a: shell shaped like a snail b: shell not shaped like a snail	Go to iii) Go to ii)
ii)	a: white shell b: black shell	Type P Type Q
iii)	a: shell with stripes or lines b: shell without stripes or lines	Go to iv) Go to v)
iv)	a: black stripes or lines b: brown stripes or lines	Type R Type S
v)	a: shell with square markings b: shell without square markings	Type T Type U

According to the table, what type of shells are the two shells circled in the photo?

- (A) Type S
- (B) Type T
- (C) Type R
- (D) Type U

Q 23. Metals expand when heated.

Jacinta had four metal rings of the same size. Each ring was made of a different metal. She wanted to see which ring would expand the most when left out in the hot sunlight.



What does Jacinta have to do to make this a fair test?

- (A) test each ring on a different day
- (B) make the mass of each ring the same
- (C) keep each ring in the sunlight for the same time
- (D) keep each ring in a different location in the sunlight

Q 24. Finn's car was leaking one drop of oil every second. The oil left a pattern of drops on the carpark.

He measured the distance between the drops of oil and drew this pattern on paper. One centimetre (cm) on paper equals one metre (m) in real life.



How was the car travelling from the start point based on the pattern of drops on the carpark?

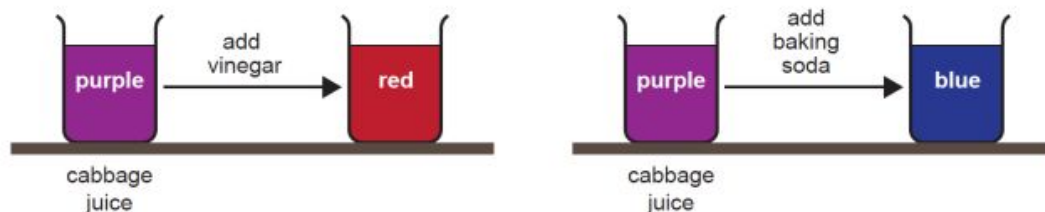
- (A) It was speeding up.
- (B) It was slowing down.
- (C) It stayed at the same speed.
- (D) It was speeding up then stopped suddenly.

Q 25. How far did the car travel in the first 6 seconds?

- (A) 6 cm
- (B) 8.5 cm
- (C) 6 m
- (D) 8.5 m

Q 26. Purple cabbage juice can be used to indicate whether a substance is alkaline or acidic.

Max took two cups of purple cabbage juice. He then added vinegar to one and baking soda to the other and noted the colour changes.



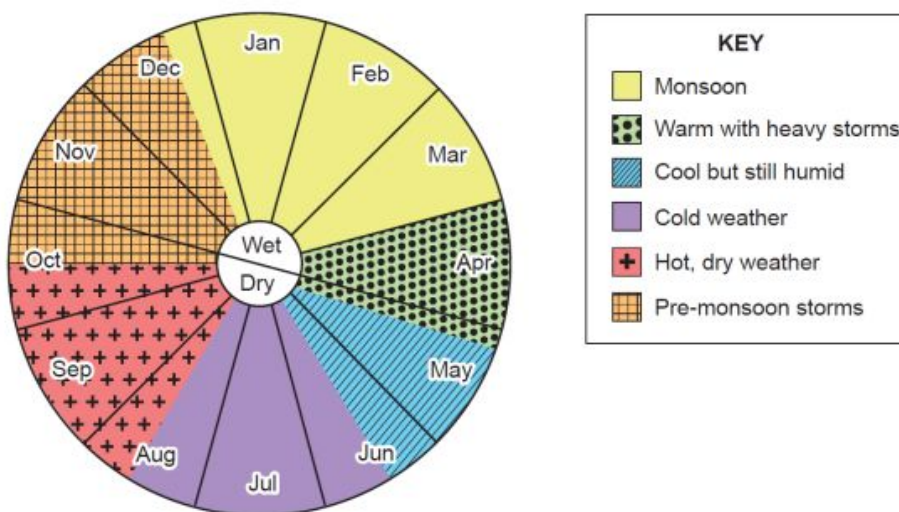
Max knew that vinegar is acidic and baking soda is alkaline. Max added some liquid he found in the kitchen to a new cup of cabbage juice. The cabbage juice turned red.

What can Max conclude from his experiment?

- (A) The cabbage juice was acidic.
- (B) The liquid contained vinegar.
- (C) The liquid was acidic.
- (D) The liquid was red.

Q 27. Aboriginal people in Australia used fire to help them manage the land. They burnt the woodlands when the weather was cool and dry. Carefully managed small and regular burns significantly reduced the intensity of the fires started by lightning during the dry weather.

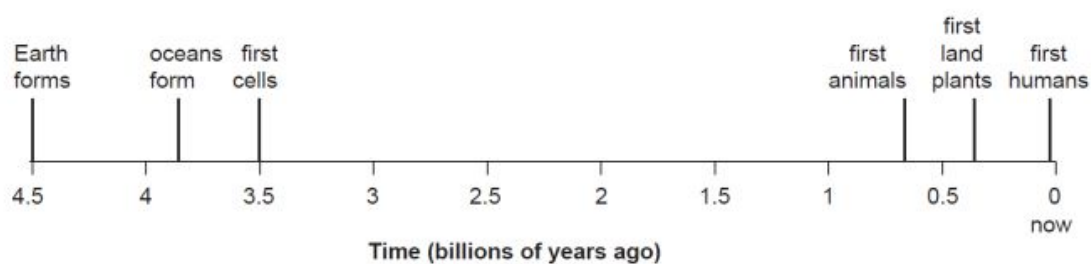
The calendar below shows the six different seasons of a year in the Kakadu region in northern Australia.



In what months did Aboriginal people in Kakadu use fire to manage the land?

- (A) February – May
- (B) June – August
- (C) August – November
- (D) November – February

Q 28. This geological time scale shows the order in which some events occurred on Earth.



Which statement about Earth's history is supported by the time scale?

- (A) The oceans started forming about 4.5 billion years ago.
- (B) The first land plants appeared on Earth before the first animals.
- (C) Humans appeared on Earth very recently.
- (D) The first animals appeared on Earth about 3.5 billion years ago.

Q 29. The table lists the times of sunrise and sunset in Sydney over seven days during May 2017.

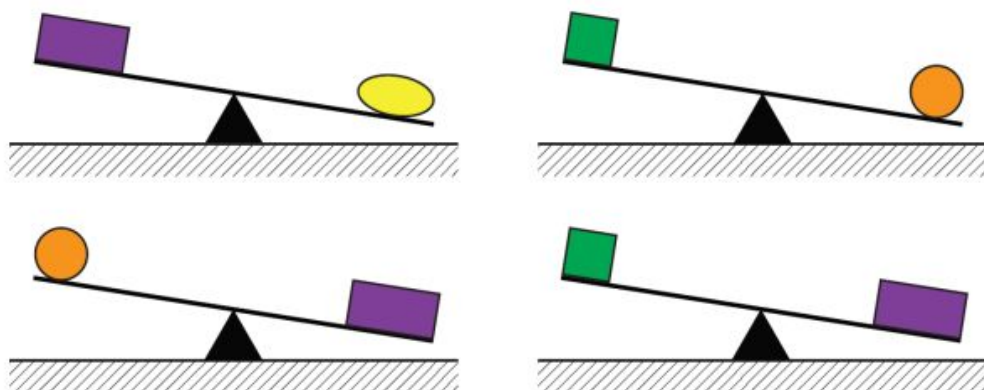
Date	Sunrise (am)	Sunset (pm)
May 21	6:44	4:58
May 22	6:45	4:58
May 23	6:45	4:57
May 24	6:46	4:57
May 25	6:47	4:56
May 26	6:47	4:56
May 27	6:48	4:55

Based on the table, what were most likely the times for sunrise and sunset on May 28, 2017?

	Sunrise (am)	Sunset (pm)
(A)	6:49	4:55
(B)	6:49	4:56
(C)	6:48	4:56
(D)	6:47	4:55

Q 30. Andy designed an experiment to determine which of the four objects was the heaviest.

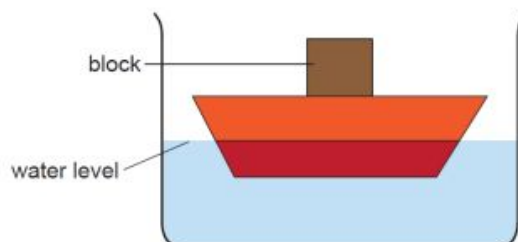
The diagrams show what Andy did to determine the heaviest object.



Which object is the heaviest?

- (A)  (B)  (C)  (D) 

Q 31. John had a wooden boat with a block on top of it, floating on water in a tank.

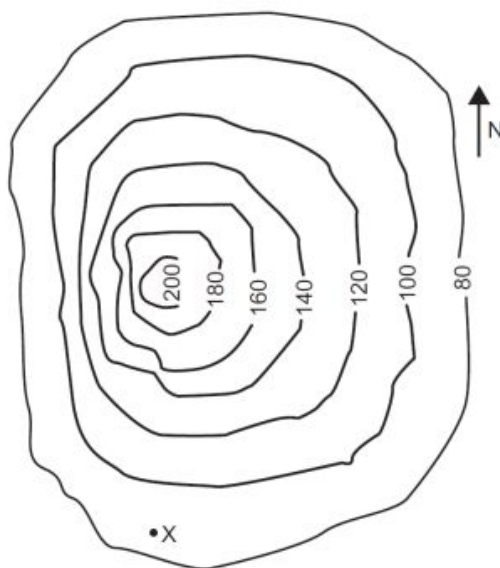


What happened to the water level when John took the block off the boat?

- (A) It went up.
 (B) It went down.
 (C) It did not change.
 (D) First it went up then it went down.

- Q 32. Lines on a contour map join the points that are at the same height above sea level. The number on each contour line is the height of all the points on the line in metres above sea level.

This contour map represents a hill.



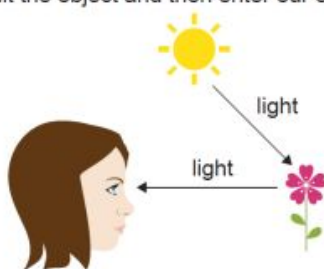
Approximately how many metres above X is the top of the hill?

- (A) 90 (B) 110 (C) 200 (D) 280
- Q 33. Dan is standing at point X looking towards the north.

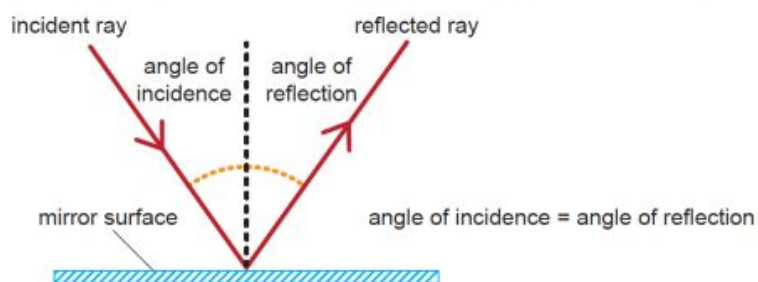
What does the hill look like from where he is standing?



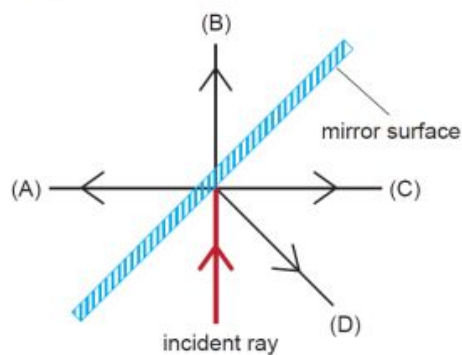
Q 34. We see an object when light rays hit the object and then enter our eyes.



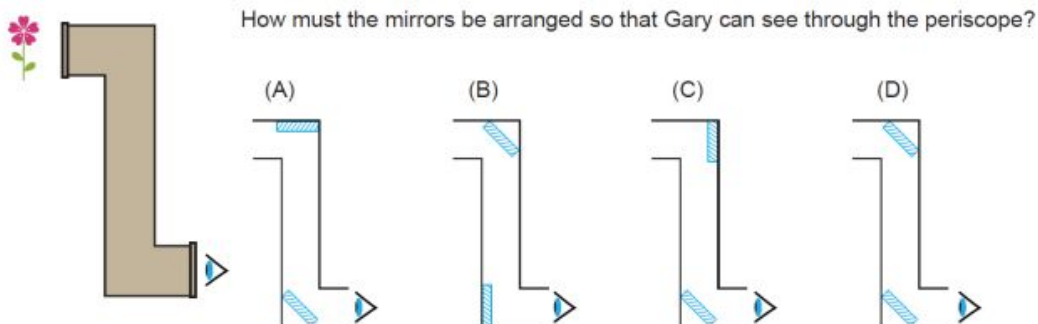
When a light ray hits a mirror surface at an angle, it will reflect off the surface and change its direction, as shown.



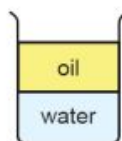
Along which arrow is the incident ray reflected?



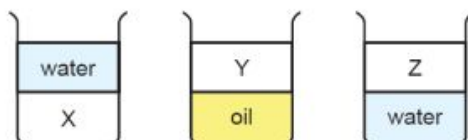
Q 35. Gary made a periscope using some pipes and two mirrors.



- Q 36. When two liquids do not mix, they separate into layers with the less dense liquid on top of the more dense one. For example, water and oil do not mix and oil is less dense than water.



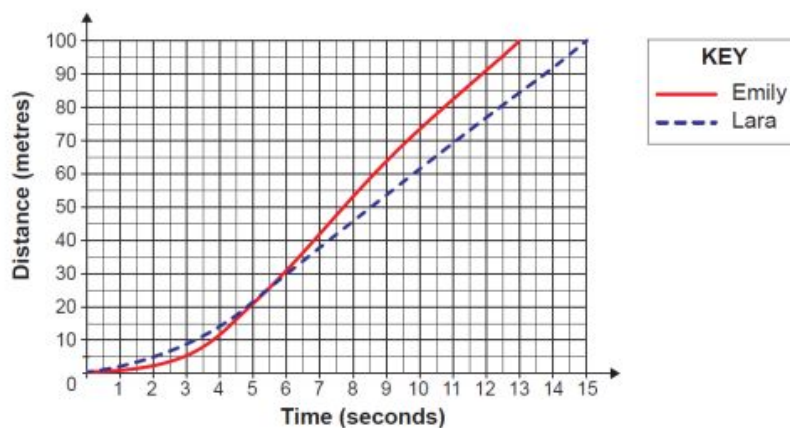
Three different liquids X, Y and Z do not mix with water or oil. The diagram shows what happens when each liquid is added to water or oil.



Which statement is supported by this information?

- (A) Z is less dense than Y.
- (B) X is more dense than Y and Z.
- (C) X and Z are as dense as water.
- (D) Y and Z are more dense than oil.

- Q 37. Emily and Lara ran a 100-metre race. The graph represents the distance over time during the race.



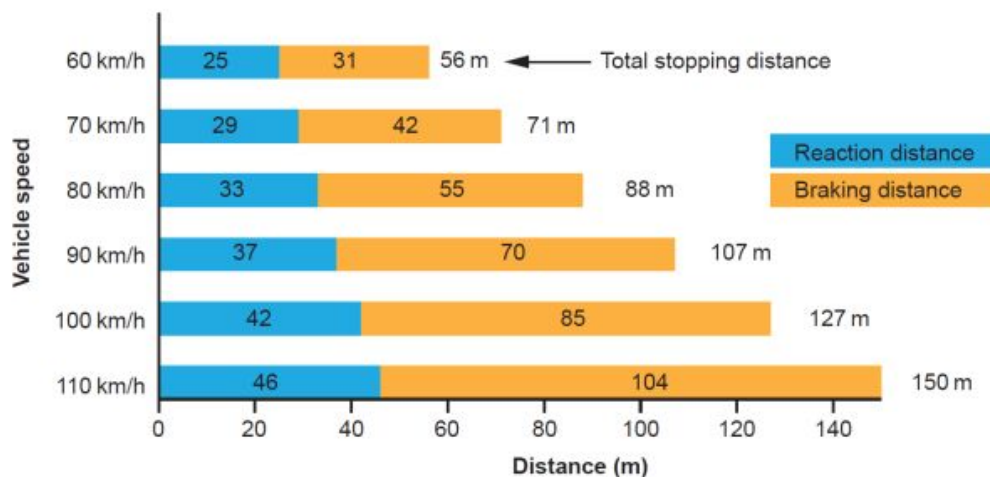
Who finished first and how many metres apart were they when the first person reached the finish line?

	Who finished first?	How far apart were they? (metres)
(A)	Emily	2
(B)	Lara	15
(C)	Emily	15
(D)	Lara	2

Q 38. The following are some factors that influence whether a driver stops in a short enough distance to avoid an accident.

- reaction distance = initial speed of the vehicle $\times$ driver's reaction time
- braking distance = the distance travelled during braking
- stopping distance = reaction distance + braking distance

The graph shows data relating to these factors.



Based on the data, which statement is correct?

- (A) The faster the vehicle is travelling, the shorter the reaction time.
 (B) Reaction distance decreases as stopping distance increases.
 (C) The slower the reaction time, the shorter the stopping distance.
 (D) The faster the vehicle speed, the greater the reaction distance.

Q 39. Louise is playing with a toy that involves interlocking gears. The main gear has 12 teeth and it makes 4 full turns in one minute.

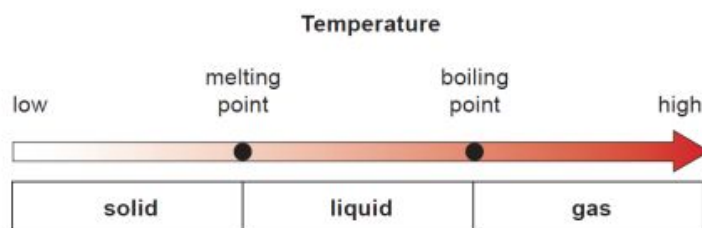


Another gear is attached to the main gear and makes 8 full turns in one minute.

How many teeth does this gear have?

- (A) 6
 (B) 8
 (C) 16
 (D) 24

- Q 40. Things around us are made up of matter. Matter can exist in three main states: solid, liquid or gas. Matter may change state as its temperature changes. The temperature at which a solid becomes a liquid is called its melting point. The temperature at which a liquid becomes a gas is called its boiling point.



The table shows the melting points and the boiling points of three substances X, Y and Z.

Substance	Melting point (°C)	Boiling point (°C)
X	10	95
Y	40	150
Z	160	320

Which statement about these substances is supported by the data in the table?

- (A) At 30 °C, X is a solid and Y is a liquid.
- (B) At 300 °C, Y is a liquid and Z is a gas.
- (C) Y and Z are gases above 160 °C.
- (D) X and Y are liquids at 45 °C.

ANSWERS

Question Number	Area Assessed	Description	Correct Answer
1	Interpreting	Interpret a diagram of a plant life cycle	A
2	Observing/Measuring	Predict the change in a shadow caused by a change in the Sun's position	A
3	Observing/Measuring	Recognise the difference between an observation, an inference and a prediction	D
4	Interpreting	Interpret a column graph	D
5	Interpreting	Match a circuit diagram with a built circuit	A
6	Investigating	Measure the difference in volume	C
7	Predicting/Concluding	Draw a conclusion from a Venn diagram	C
8	Observing/Measuring	Estimate a length by comparison	B
9	Investigating	Determine a way to carry out an investigation	A
10	Observing/Measuring	Recognise differences between properties of objects and properties of materials	B
11	Reasoning/Problem solving	Predict a likely outcome following a change in a simple food chain	B
12	Reasoning/Problem solving	Solve a problem by spatial reasoning	A
13	Interpreting	Locate information in a table	D
14	Predicting/Concluding	Identify the statement supported by data provided in a table	B
15	Predicting/Concluding	Draw a conclusion from information in a table	C
16	Interpreting	Interpret an unfamiliar graph	B
17	Investigating	Identify the dependent variable in an investigation	C
18	Investigating	Identify a variable that needs to be kept constant	D
19	Interpreting	Interpret an Earth-Sun diagram	D
20	Observing/Measuring	Read a thermometer with dual scales	C

21	Investigating	Identify the reason for a step in an investigation	C
22	Interpreting	Use a dichotomous key to classify some objects	D
23	Investigating	Recognise the conditions for an investigation to be a fair test	C
24	Predicting/Concluding	Predict an outcome based on a pattern	A
25	Observing/Measuring	Measure distance using a scale	D
26	Investigating	Draw a conclusion based on the results of an investigation	C
27	Interpreting	Interpret a diagram of an Aboriginal calendar	B
28	Interpreting	Interpret a geological time scale	C
29	Reasoning/Problem solving	Identify patterns in tabulated data and make a prediction	A
30	Predicting/Concluding	Determine the relative masses of objects using a balance	D
31	Predicting/Concluding	Predict the outcome of a change to a system	B
32	Interpreting	Interpret a topographic map to determine the height of a hill	B
33	Predicting/Concluding	Identify landforms represented by a contour map	B
34	Predicting/Concluding	Identify the correct path of a reflected ray	C
35	Reasoning/Problem solving	Determine the structure of a device based on its function	D
36	Reasoning/Problem solving	Determine relationships by logical comparison	B
37	Predicting/Concluding	Draw conclusions from a line graph	C
38	Reasoning/Problem solving	Identify a relationship based on data in a graph	D
39	Reasoning/Problem solving	Solve a problem about a gear system	A
40	Reasoning/Problem solving	Determine the states of substances based on given information	D